

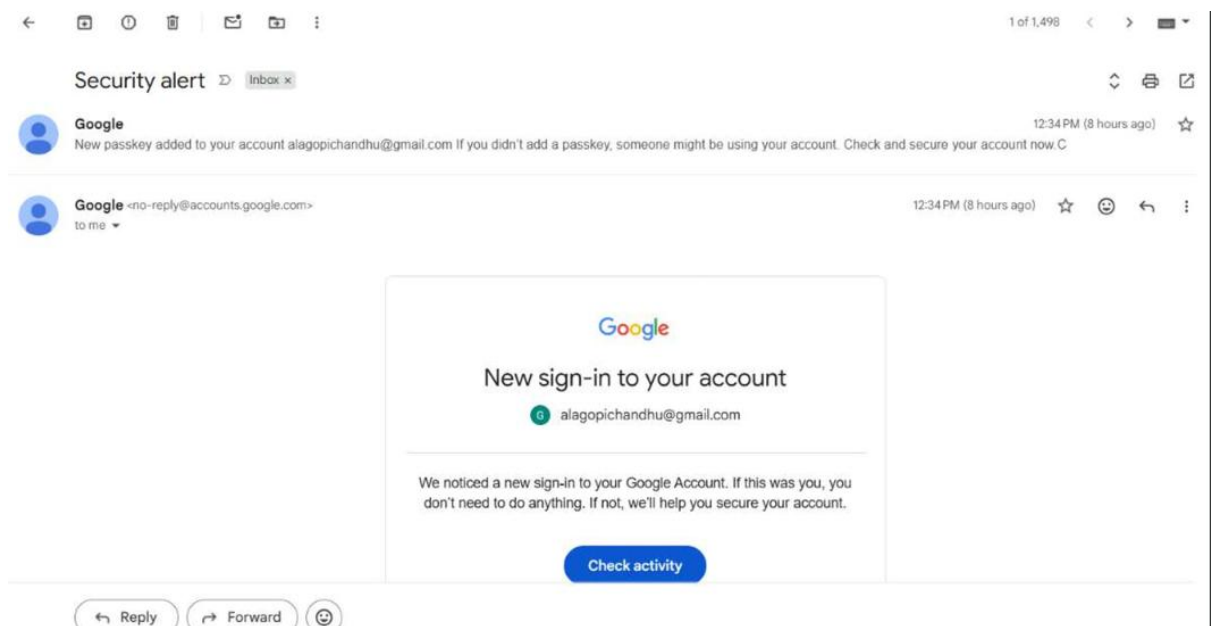
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.

- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

Original Message

Message ID	<OUQKH5g1e9U4Jb56H7G4Cg@notifications.google.com>
Created at:	Wed, Aug 26, 2026 at 12:34 PM (Delivered after 2 seconds)
From:	Google <no-reply@accounts.google.com>
To:	alagopichandhu@gmail.com
Subject:	Security alert
SPF:	PASS with IP 209.85.220.73 Learn more
DKIM:	'PASS' with domain accounts.google.com Learn more
DMARC:	'PASS' Learn more

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```
Delivered-To: alagopichandhu@gmail.com
Received: by 2002:a05:6359:2706:b0:2b2:72f4:d5db with SMTP id lw6csp6345786rwb; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
X-Received: by 2002:a05:690c:e1cb:20b0:7fd:a7b6:8d87 with SMTP id 06721157ae682-857486b068mr16366277b3.25.1787727864286; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787727864; cv=none; d=google.com; s=arc-20260327; b=1Pko9gC0A0H5aL11fR3BP89dHqMn/yov8/agmV4rJ8Q/6Zxfta9xCa8LJv9pVKhT
vMAqyBBB1gHaUm55XE5PMsvoh470x0dJHte34k7pUmoqm10pR/1Com07Ga1H/Um03Eya GGDaENjMTz9/GXvyZ2sup7y+s6YPsD/VQqpahvaIW/D3DETMrsVpmJc7AYgicilB70Ch
dfrwHhX2dIT1WypAmEwZfCcxANTMPHDOwv558XbH1b0X61bo3FADQ/xF2hjOGdlrgTzVT WORE/5+5eNTH8tw0eCR7j5FroHQ4Iaesxq7XvzTiPoPRg3XdShi/ytmOeEmFM/555+6 fqnA==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=to:from:subject:message-id:globalsai:feedback-id:date:mime-
```

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

Pricing
Tools
Delivery Center
Monitoring
Products
Resources
Login

SuperTool
MX Lookup
Blacklists
DMARC
Diagnostics
Email Health
DNS Lookup
Analyze Headers
All Tools

Email Header Analyzer

Paste Header:

```
Delivered-To: alagopichandhu@gmail.com
Received: by 2002:a05:6359:2706:b0:2b2:72f4:d5db with SMTP id lw6csp6345786rwb; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
X-Received: by 2002:a05:690c:e1cb:20b0:7fd:a7b6:8d87 with SMTP id 00721157ae682-857486b068mr16366277b3.25.1787727864286; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787727864; cv=none; d=google.com; s=arc-20260327; b=1Pko9gC0A0H5aL11fR3BP89dHqMn/yov8/agmV4rJ8Q/6Zxfta9xCa8LJv9pVKhT
vMAqyBBB1gHaUm55XE5PMsvoh470x0dJHte34k7pUmoqm10pR/1Com07Ga1H/Um03Eya GGDaENjMTz9/GXvyZ2sup7y+s6YPsD/VQqpahvaIW/D3DETMrsVpmJc7AYgicilB70Ch
dfrwHhX2dIT1WypAmEwZfCcxANTMPHDOwv558XbH1b0X61bo3FADQ/xF2hjOGdlrgTzVT WORE/5+5eNTH8tw0eCR7j5FroHQ4Iaesxq7XvzTiPoPRg3XdShi/ytmOeEmFM/555+6 fqnA==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=to:from:subject:message-id:globalsai:feedback-id:date:mime-version:dkim-signature;
b=1Pko9gC0A0H5aL11fR3BP89dHqMn/yov8/agmV4rJ8Q/6Zxfta9xCa8LJv9pVKhT vMAqyBBB1gHaUm55XE5PMsvoh470x0dJHte34k7pUmoqm10pR/1Com07Ga1H/Um03Eya GGDaENjMTz9/GXvyZ2sup7y+s6YPsD/VQqpahvaIW/D3DETMrsVpmJc7AYgicilB70Ch
dfrwHhX2dIT1WypAmEwZfCcxANTMPHDOwv558XbH1b0X61bo3FADQ/xF2hjOGdlrgTzVT WORE/5+5eNTH8tw0eCR7j5FroHQ4Iaesxq7XvzTiPoPRg3XdShi/ytmOeEmFM/555+6 fqnA==
vWwAqz9Dio236eUjCQ7B0bJCKDKCoG+Az5baXg3Cf5n81buaK8VpJaQhVbQTSJEfKo bmlk9CnTzxiQpnfUpjx0X2iXZGQV/S0dR5cDNkS6DuxZoeqR00/rzBA+jymOU uDAQ==; dara=google.com
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, just [read this tutorial](#).

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.

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- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

The screenshot shows the MXToolbox Header Analyzer interface. The email subject is "Security alert". Under "Delivery Information", all checks (DMARC, SPF, DKIM) are marked as compliant or aligned. Under "Relay Information", the received delay is 0 seconds. A diagram shows the email path from mail-scr73.google.com to mx.google.com.

7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

The screenshot shows the "Relay (Seconds)" table and "SPF and DKIM Information".

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail-scr73.google.com 209.85.220.73	mx.google.com	SMTPS	8/26/2026 7:04:24 AM	✓
2	0 seconds		2002:a05:6359:2706:b0:2b2:7294:d6db	SMTP	8/26/2026 7:04:24 AM	

SPF and DKIM Information:

- DMARC: accounts.google.com (v=DMARC1; p=reject; rua=mailto:mailauth-reports@google.com)
- SPF: gaia.bounces.google.com:209.85.220.73

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Solve Email Delivery Problems

Show

```
v=1, m=rsa-sha256; c=relaxed/relaxed; d=accounts.google.com; s=20251104; t=1707727064; x=1708332664; dara=google.com; h=content-type; to=from:subject:message-id:mail-feedback-id:date :mime-ver
```

[illegible]

```
Delivered-To: algojichandhu@gmail.com
Received: by 2002:a95:659e:720c:b0:2b2:72f4:de6d with SMTP id lufscsp545786rwh; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
X-Received: by 2002:a95:659e:04c1:20b9:7fd:a7b5:d857 with SMTP id 00721157ae682-85736b29bafmr2121216763.6.1787272863954; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
ARC-Seal: i=1; arcsha256=:t=1787272864; c=v-mee; d=google.com; s=wrc-20250327; b=LPl9sgdQk0BdlHlFJ3FB9d4gW/yJoVb/agnVnA=78Q/Cxfta0CaLLjy9vYVht-vMAYz0D0GflWak5XXSF5HVuh476n3dJHTe34kjJdHqon0ghU/
ARC-Message-Signature: i=1; arcsha256=:carelated/relaxed; d=google.com; saarc-20260327; hoto:from:subject:message-id:gmail:feedback-id:date:mime-version :dkim-signature; bh=8xc6RZ5da27FFQd/rntspFdId
ARC-Authentication-Results: i=1; sha256; dkimpass header.i@accounts.google.com hsdor.s=20251104 header.b=ml19345; spfpass (google.com: domain of 394-oaggtqdzwa-dqbckmoazgrfe.czaosq.oaymensuboutptgzmtygs.oay@gaii.bounces.google.com)
Return-Path: <394-OaggtDwZa=dQLXaVOOmg2Fe.SaaSQ.OaVM00saLUOTN2PI65PMUX.OaY@gaii.bounces.google.com>
Received: from mail-sor-f73.google.com (mail-sor-f73.google.com [209.85.220.73]) by mx.google.com with SMTPS id 00721157ae682-857802344dbiscr24685457b3.13.2026.08.26.00.04.24 for <algojichandhu@gmail.
Received-SPF: pass (google.com: domain of 394-oaggtqdzwa-dqbckmoazgrfe.czaosq.oaymensuboutptgzmtygs.oay@gaii.bounces.google.com designates 209.85.220.73 as permitted sender) client-ip=209.85.220.73;
Authentication-Results: mx.google.com; dkimpass header.i@accounts.google.com header.s=20251104 header.b=hlTc345; spfpass (google.com: domain of 394-oaggtqdzwa-dqbckmoazgrfe.czaosq.oaymensuboutptgzmtygs.oay@gaii.bounces.google.com)
DKIM-Signature: v=1; arcsha256=:carelated/relaxed; d=accounts.google.com; s=20251104; t=1787272864; x=1788326664; daramgoogle.com; h=content-type:from:subject:message-id:gmail:feedback-id:date:m
X-Google-DKIM-Signature: v=1; arcsha256=:carelated/relaxed; d=icloud.net; s=20251104; t=1787272864; x=1788326664; h=content-type:from:subject:message-id:gmail:feedback-id:date:mime-version;x-gem
X-Gm-Message-State: Afu+nfFnDUpsixtkhsSc7eryd3oVti2bzKOXXoeRtCioChfDmP=s20klou5MPH/Cx3aOU6ta5c1e3Y5vtYztzf4Y2o2a3xc000HAduItonKsfadWad7API/Sbo0/WII2ow=s0tn8glCmf5Ctoi405SVU7YUsr/DN-
X-MIME-Version: 1.0
X-Received: by 2002:a95:659e:6082:b0:845:2054:2357 with SMTP id 00721157ae682-85736b29bafmr2121216763.6.1787272863954; Wed, 26 Aug 2026 00:04:24 -0700 (PDT)
Date: Wed, 26 Aug 2026 07:04:22 GMT
X-Account-Notification-Type: 325
Feedback-ID: 325:account-notification
gmail: true
X-Google-Notification-Metadata: CjTSAA
X-Notifications: b4f9aef49a9000
X-Notifications-Source-Info: ANkoAQCGd4Mc-L0q2tle-PCtlMs4qcKm6WMIDvSP7DXtIXly1ZkaobgdjRu2f4fRetLoSOcrBmVch3cJs5uW9fR7DVCvJB03JJ7Khovfy24f0Q0G0Ur-i-aPfBV4odH2bVsC_k_n_PCS9J8CR2ev6WTii1T335
Message-ID: <COUQ9YisR6u356H7GAC@gmail-notifications.google.com>
Subject: Security alert
From: Google <no-reply@accounts.google.com>
To: algojichandhu@gmail.com
Content-Types: multipart/alternative; boundary="0000000000000493bf0F5edd9dc"

--0000000000000493bf0F5edd9dc
Content-Type: text/plain; charset="UTF-8"; format=flowed; delsp=yes
Content-Transfer-Encoding: base64
```

[illegible]